General Chemistry (Chem1006)

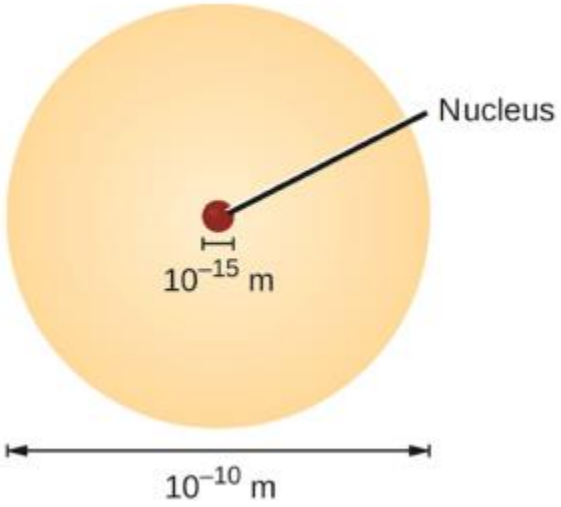
Chapter 2

Atoms, Molecules and Ions

2.1. Atomic structure and symbolism

An atom contains:

- a very small **nucleus** composed of positively charged protons and uncharged **neutrons**,
- surrounded by a much larger volume of space containing negatively charged **electrons**.
- The nucleus contains the **majority of an atom's mass** because protons and neutrons are much heavier than electrons,
- Electrons occupy almost all of **an atom's volume**.



- ✓ we use appropriately small units of measure, such as the atomic mass unit (amu) and the fundamental unit of charge (e).
  one amu is exactly 1/12 of the mass of one carbon-12 atom:

  $1\ \text{amu} = 1.6605 \times 10^{-24}\ \text{g}$.
- ✓ The fundamental unit of charge equals the magnitude of the charge of an electron (e) with **$e = 1.602 \times 10^{-19}\ \text{C}$.**

Table2.1: Properties of Subatomic Particles

Name	Location	Charge (C)	Unit Charge	Mass (amu)	Mass (g)
Electron	outside nucleus	-1.602×10^{-19}	1−	0.00055	0.00091×10^{-24}
Proton	Nucleus	1.602×10^{-19}	1+	1.00727	1.67262×10^{-24}
Neutron	Nucleus	0	0	1.00866	1.67493×10^{-24}

- The number of protons in the nucleus of an atom is its **atomic number (Z).**
- The total number of protons and neutrons in an atom is called its **mass number (A)**

$$atomic\ number(Z) = number\ of\ protons$$

$$mass\ number(A) = number\ of\ protons + number\ of\ neutrons$$

$$A - Z = number\ of\ neutrons$$

- When the numbers of these subatomic particles are not equal, the atom is electrically charged and is called **an ion**. The charge of an atom is defined as follows:

$$\text{Atomic charge} = \text{number of protons} - \text{number of electrons}$$

- An atom that gains one or more electrons will exhibit a negative charge and is called **an anion**. Positively charged atoms called **cations** are formed when an atom loses one or more electrons.

Example: The iodine atoms are added as anions, and each has a 1− charge and a mass number of 127. Determine the numbers of protons, neutrons, and electrons in one of these iodine anions.

Solution

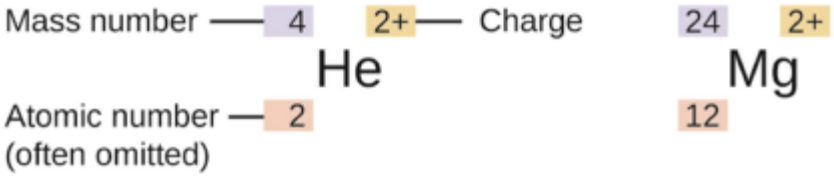
The **atomic number of iodine (53)** tells us that a neutral iodine atom **contains 53 protons** in its nucleus and **53 electrons** outside its nucleus. Because the **sum of the numbers of protons and neutrons equals** the mass number, 127, the number of neutrons is **74 (127 − 53 = 74).** Since the iodine is added as a 1− anion, **the number of electrons is 54 [53 – (1–) = 54]**.

2.1.1. Chemical Symbols and Isotopes

- **A chemical symbol** is an abbreviation that we use to indicate an element or an atom of an element. For example, the symbol for mercury is Hg.
  - Most symbols have one or two letters,
  - Only the first letter of a symbol is capitalized. For example, **Co is the symbol for the element cobalt**. **CO is for carbon-monoxide**.

- **Isotopes:** are members of a family of an element that all have the same number of protons but **different numbers of neutrons**.

For example, magnesium exists as a mixture of three isotopes, each with an atomic number of 12 and with mass numbers of 24, 25, and 26, respectively. These isotopes can be identified as **^{24}Mg, ^{25}Mg, and ^{26}Mg**.



Element	Symbol	Atomic No.	No of Protons	No of Neutrons	Mass (amu)	% Natural Abundance
hydrogen	$^{1}_{1}H$(protium)	1	1	0	1.0078	99.989
	$^{2}_{1}H$(deuterium)	1	1	1	2.0141	0.0115
	$^{3}_{1}H$(tritium)	1	1	2	3.01605	— (trace)
Helium	$^{3}_{2}He$	2	2	1	3.01603	0.00013
	$^{4}_{2}He$	2	2	2	4.0026	100
Oxygen	$^{16}_{8}O$	8	8	8	15.9949	99.757
	$^{17}_{8}O$	8	8	9	16.9991	0.038
	$^{18}_{8}O$	8	8	10	17.9992	0.205

2.1.2. Atomic mass unit and average atomic mass

- The atomic mass of a single atom is approximately equal to its mass number (a whole number).
- The average masses of atoms of most elements are not whole numbers because most elements exist naturally as mixtures of two or more isotopes.

$$\text{average mass} = \sum_i (\text{fractional abundance} \times \text{isotopic mass})_i$$

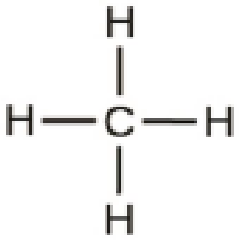
For example, the element boron is composed of two isotopes: About 19.9% of all boron atoms are ^{10}B with a mass of 10.0129 amu, and the remaining 80.1% are ^{11}B with a mass of 11.0093 amu. The average atomic mass for boron is calculated to be:

$$boron\ average\ mass = (0.199 \times 10.0129\ amu) + (0.801 \times 11.0093\ amu)$$
$$= 1.99\ amu + 8.82\ amu = 10.81\ amu$$

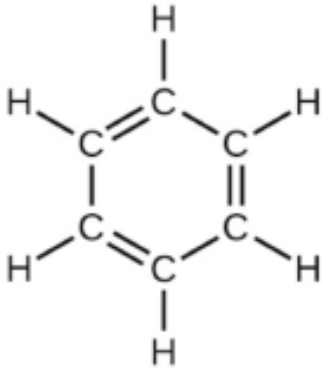
2.2. Chemical Formulas

- **A molecular formula** is a representation of a molecule that **uses chemical symbols** to indicate the types of atoms followed by **subscripts t**o show the number of atoms of each type in the molecule.
- **The structural formula** for a compound gives the same information as its molecular formula but also shows how the atoms are connected in the molecule.

CH_4



C_6H_6



- **Empirical Formula**: is the **simple whole number ratio of atoms** present in a compound.

For example, the molecular formula for acetic acid is $C_2H_4O_2$. The ratio of atoms is 2:4:2. Dividing by the lowest common denominator (2) gives the simplest, whole-number ratio of atoms, 1:2:1, so the empirical formula is CH_2O.

2.3. The Periodic Table

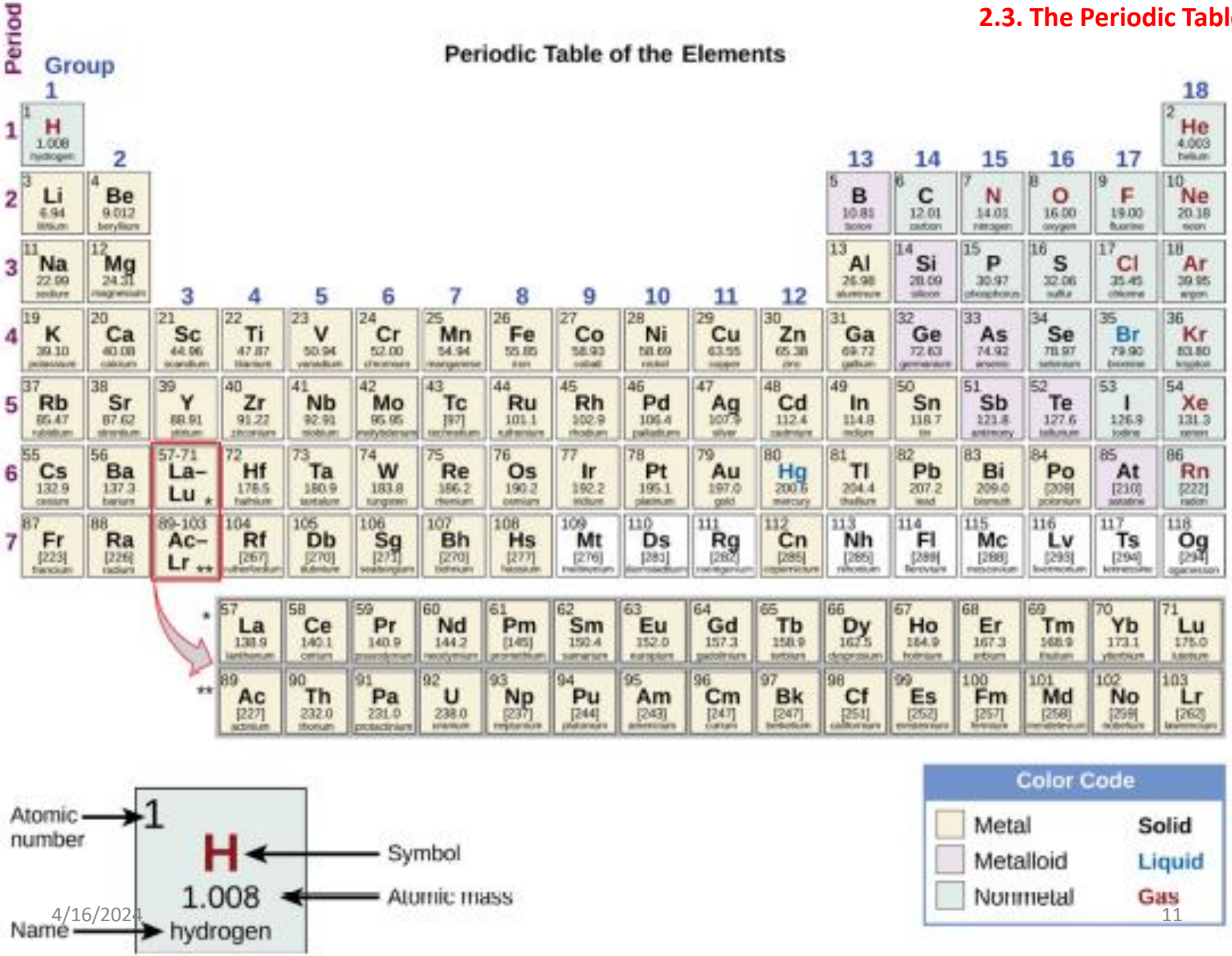
2.3.1. Historical development of the periodic table

- **Dimitri Mendeleev** in Russia (1869) and **Lothar Meyer** in Germany (1870) **independently recognized that there was a periodic relationship** among the properties of the elements known at that time. Both published tables with the elements arranged according to **increasing atomic mass**.

Mendeleev's periodic table

Reihen	Gruppe I. — R^1O	Gruppe II. — RO	Gruppe III. — R^2O^3	Gruppe IV. RH^4 RO^2	Gruppe V. RH^3 R^2O^5	Gruppe VI. RH^2 RO^3	Gruppe VII. RH R^2O^7	Gruppe VIII. — RO^4
1	H=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,3	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Cu=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Sr=87	?Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ru=104, Rh=104, Pd=106, Ag=108.
7	(Ag=108)	Cd=112	In=113	Sn=118	Sb=122	Te=125	J=127	
8	Cs=133	Ba=137	?Di=138	?Ce=140	—	—	—	— — — —
9	(—)	—	—	—	—	—	—	
10	—	—	?Er=178	?La=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	(Au=199)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	— — — —

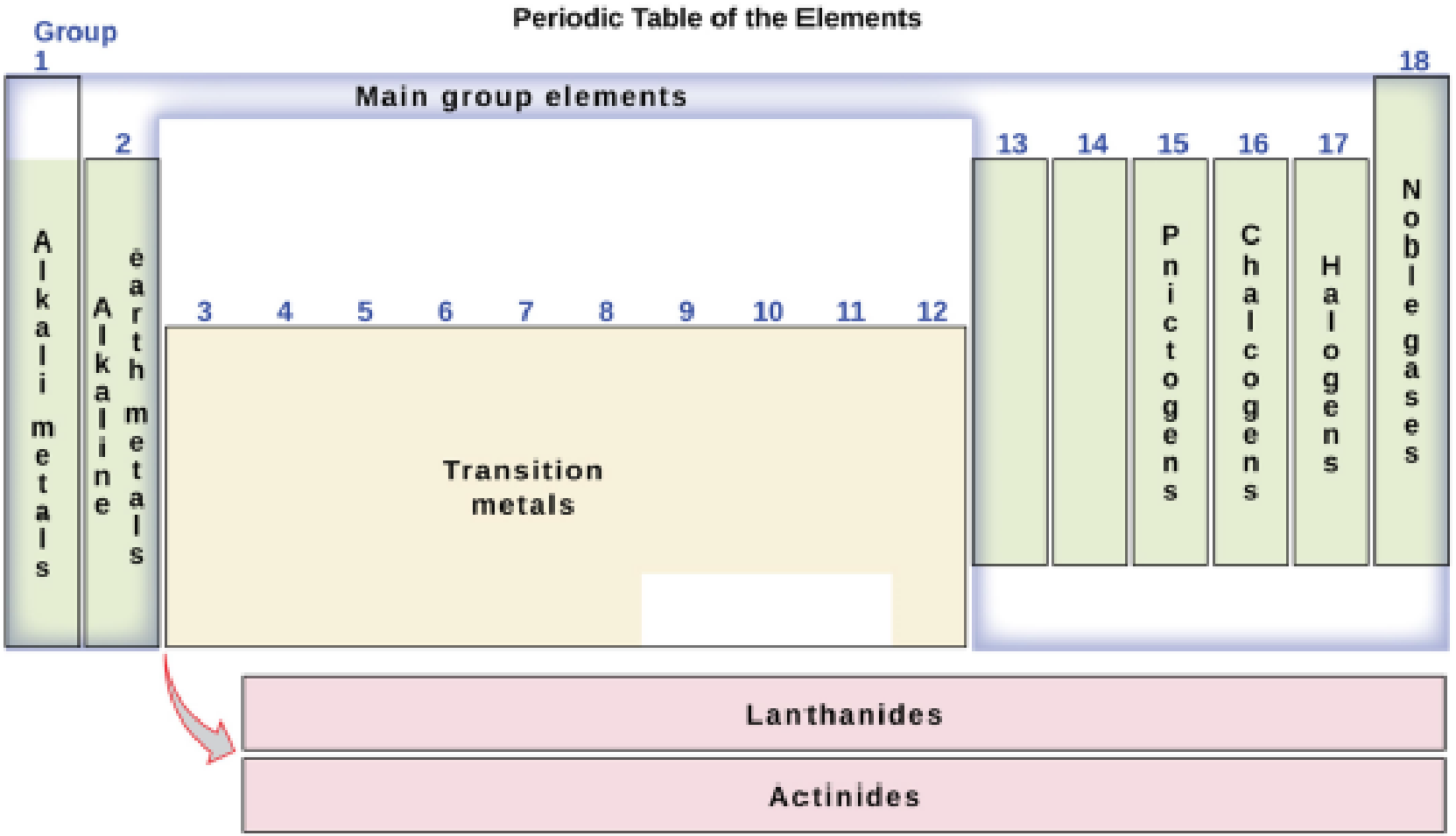
- By the twentieth century, it became apparent that the **periodic relationship** involved **atomic numbers** rather than atomic masses. The modern statement of this relationship, the periodic law, is as follows:
  - ✓ The properties of the elements are periodic functions of their atomic numbers.
  - ✓ A modern periodic table arranges the elements in increasing order of their atomic numbers and groups atoms with similar properties in the same vertical column.
  - ✓ Each box represents an element and contains its atomic number, symbol, average atomic mass, and (sometimes) name.
  - ✓ The elements are arranged in seven horizontal rows, called periods or series, and 18 vertical columns, called groups.


Periodic Table of the Elements
Period
Group
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
H 1.008 hydrogen
He 4.003 helium
Li 6.94 lithium
Be 9.012 beryllium
B 10.81 boron
C 12.01 carbon
N 14.01 nitrogen
O 16.00 oxygen
F 19.00 fluorine
Ne 20.18 neon
Na 22.99 sodium
Mg 24.31 magnesium
Al 26.98 aluminium
Si 28.09 silicon
P 30.97 phosphorus
S 32.06 sulfur
Cl 35.45 chlorine
Ar 39.95 argon
K 39.10 potassium
Ca 40.08 calcium
Sc 44.96 scandium
Ti 47.87 titanium
V 50.94 vanadium
Cr 52.00 chromium
Mn 54.94 manganese
Fe 55.85 iron
Co 58.93 cobalt
Ni 58.69 nickel
Cu 63.55 copper
Zn 65.38 zinc
Ga 69.72 gallium
Ge 72.63 germanium
As 74.92 arsenic
Se 78.97 selenium
Br 79.90 bromine
Kr 83.80 krypton
Rb 85.47 rubidium
Sr 87.62 strontium
Y 88.91 yttrium
Zr 91.22 zirconium
Nb 92.91 niobium
Mo 95.95 molybdenum
Tc [97] technetium
Ru 101.1 ruthenium
Rh 102.9 rhodium
Pd 106.4 palladium
Ag 107.9 silver
Cd 112.4 cadmium
In 114.8 indium
Sn 118.7 tin
Sb 121.8 antimony
Te 127.6 tellurium
I 126.9 iodine
Xe 131.3 xenon
Cs 132.9 caesium
Ba 137.3 barium
57-71 La–Lu *
Hf 178.5 hafnium
Ta 180.9 tantalum
W 183.8 tungsten
Re 186.2 rhenium
Os 190.2 osmium
Ir 192.2 iridium
Pt 195.1 platinum
Au 197.0 gold
Hg 200.6 mercury
Tl 204.4 thallium
Pb 207.2 lead
Bi 209.0 bismuth
Po [209] polonium
At [210] astatine
Rn [222] radon
Fr [223] francium
Ra [226] radium
89-103 Ac–Lr **
Rf [267] rutherfordium
Db [270] dubnium
Sg [271] seaborgium
Bh [270] bohrium
Hs [277] hassium
Mt [276] meitnerium
Ds [281] darmstadtium
Rg [282] roentgenium
Cn [285] copernicium
Nh [286] nihonium
Fl [289] flerovium
Mc [288] moscovium
Lv [293] livermorium
Ts [294] tennessine
Og [294] oganesson
* La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu
** Ac Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No Lr
Atomic number
Symbol
Atomic mass
Name
Color Code
Metal
Metalloid
Nonmetal
Solid
Liquid
Gas

2.3.2. Classification of elements in the periodic table

- Many elements differ dramatically in their chemical and physical properties, but some elements are similar in their behaviors.
  - ✓ For example, many elements appear shiny, are malleable (able to be deformed without breaking) and ductile (can be drawn into wires), and conduct heat and electricity well.
  - ✓ Other elements are not shiny, malleable, or ductile, and are poor conductors of heat and electricity.
- We can sort the elements into large classes with common properties:
  - ✓ **Metals:** elements that are shiny, malleable, good conductors of heat and electricity;
  - ✓ **Nonmetals:** elements that appear dull, poor conductors of heat and electricity and
  - ✓ **Metalloids:** elements that conduct heat and electricity moderately well, and possess some properties of metals and some properties of nonmetals.
- The elements can also be classified into the
  - ✓ main-group elements (or representative elements) in the columns labeled 1, 2, and 13–18;
  - ✓ the transition metals in the columns labeled 3–12; and
  - ✓ inner transition metals in the two rows at the bottom of the table (the top-row elements are called lanthanides and the bottom-row elements are actinides;

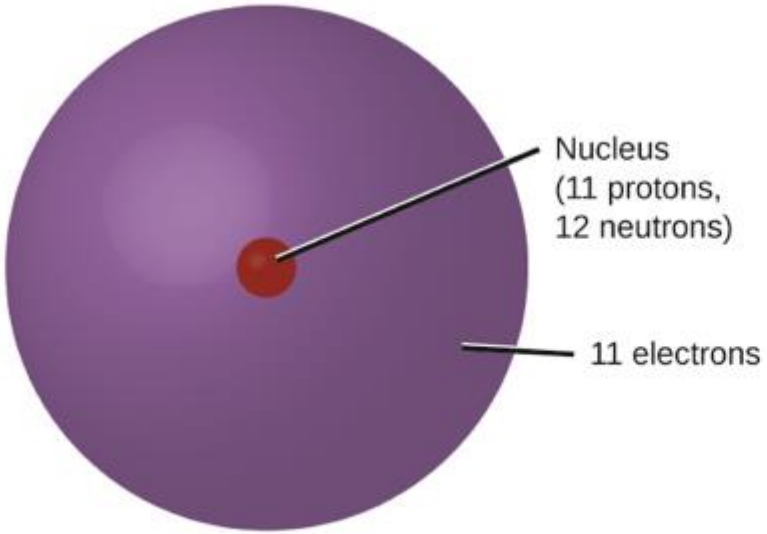
- The elements can be subdivided further by more specific properties, such as the composition of the compounds they form. For example,
  - ✓ **the elements in group 1** (the first column) form **compounds that consist of one atom of the element and one atom of hydrogen**. These elements (except hydrogen) are known as alkali metals, and they all have similar chemical properties.
  - ✓ **The elements in group 2** (the second column) form compounds consisting of **one atom of the element and two atoms of hydrogen**: These are called alkaline earth metals, with similar properties among members of that group.
  - ✓ Other groups with specific names are the pnictogens (group 15), chalcogens (group 16), halogens (group 17), and the noble gases (group 18, also known as inert gases).
- Hydrogen is a unique, nonmetallic element with properties similar to both group 1 and group 17 elements. For that reason, hydrogen may be shown at the top of both groups, or by itself.


Periodic Table of the Elements
Group 1
18
Main group elements
2
13
14
15
16
17
Alkali metals
Alkaline earth metals
3
4
5
6
7
8
9
10
11
12
Transition metals
Pnictogens
Chalcogens
Halogens
Noble gases
Lanthanides
Actinides

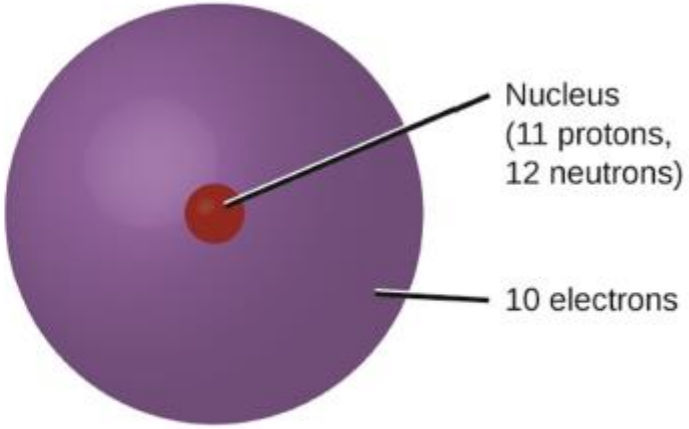
2.4. Ionic and Molecular Compounds

- In ordinary chemical reactions, the nucleus of each **atom remains unchanged**.
- Electrons, however, **can be added to atoms** by transfer from other atoms, **lost** by transfer to other atoms, or **shared** with other atoms.
- The transfer and sharing of electrons among atoms govern the chemistry of the elements.
- During the formation of some compounds**, atoms gain or lose electrons**, and form electrically charged particles called ions

Example: A sodium atom (Na) has equal numbers of protons and electrons (11) and is uncharged. A sodium cation (Na^+) has lost an electron, so it has one more proton (11) than electrons (10), giving it an overall positive charge, signified by a superscripted plus sign.



Na atom



Na^+ ion

- Atoms of many main-group metals **lose enough electrons** to leave them with the same number of electrons as an atom of the preceding noble gas.
  - ✓ To illustrate, an atom of an alkali metal (group 1) loses one electron and forms a **cation with a 1^+ charge**; an alkaline earth metal (group 2) loses two electrons and forms a **cation with a 2^+ charge**, and so on.

For example, a neutral calcium atom, with 20 protons and 20 electrons, readily loses two electrons. This results in a cation with 20 protons, 18 electrons, and a **2+ charge**. It has the same number of electrons as atoms of the preceding noble gas, argon, and is **symbolized Ca^{2+}**

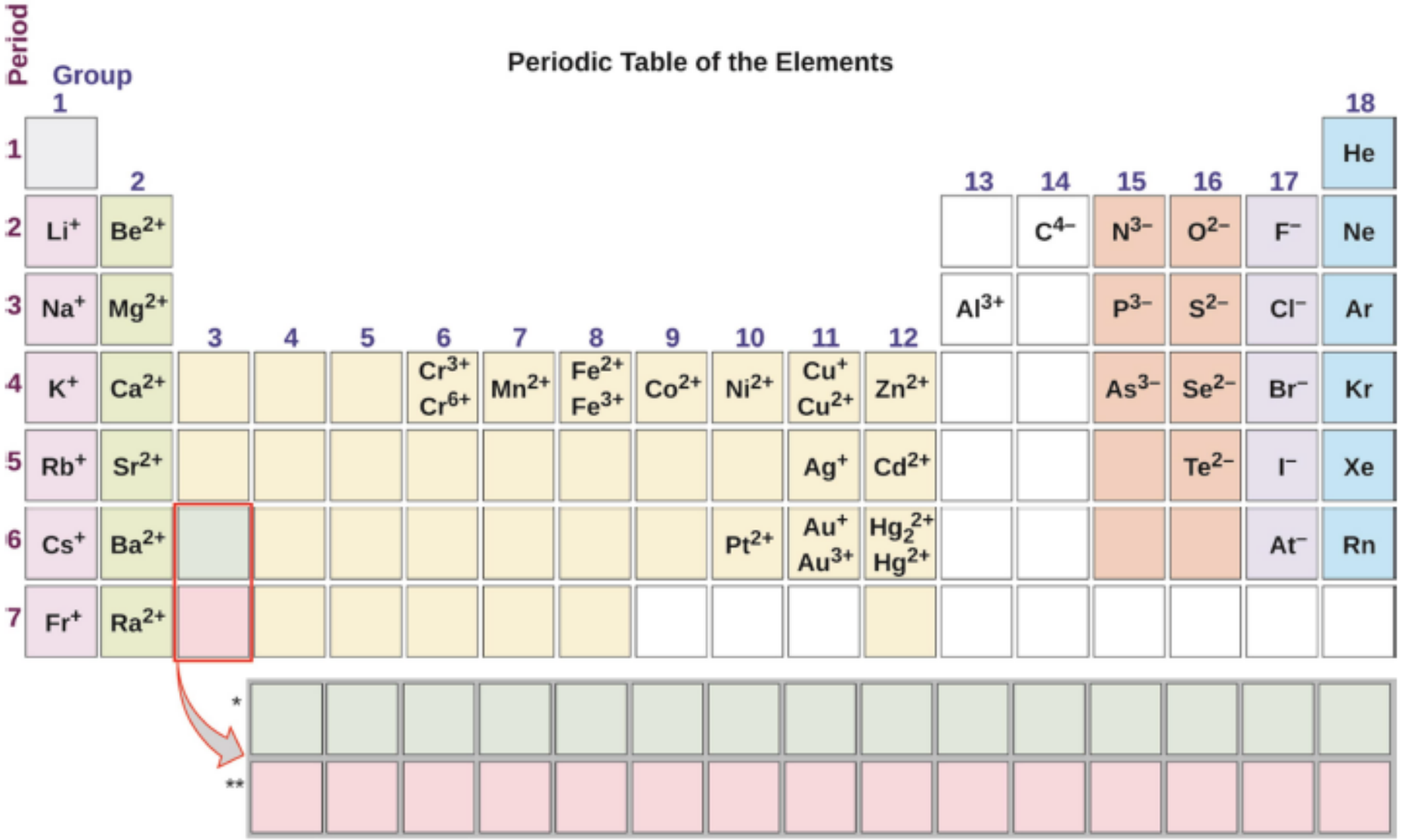
- When atoms of nonmetal elements form ions, they generally **gain enough electrons** to give them the same number of electrons as an atom of the next noble gas in the periodic table.
  - ✓ Atoms of group 17 gain one electron and form **anions with a 1– charge**; atoms of group 16 gain two electrons and **form ions with a 2– charge**, and so on.

For example, the neutral bromine atom, with 35 protons and 35 electrons, can gain one electron to provide it with 36 electrons. This results in an anion with 35 protons, 36 electrons, and a **1– charge**. It has the same number of electrons as atoms of the next noble gas, krypton, and is **symbolized Br–**.

- Moving from the **far left to the right** on the periodic table, main-group elements tend to **form cations with a charge equal to the group number**. That is, group 1 elements form 1+ ions; group 2 elements form 2+ ions, and so on.

- Moving from the **far right to the left** on the periodic table, elements often **form anions with a negative charge equal to the number of groups** moved left from the noble gases. **For example**, group 17 elements (one group left of the noble gases) form 1− ions; group 16 elements (two groups left) form 2− ions, and so on.

- Transition metals and some other metals often exhibit **variable charges** that are not predictable by their location in the table. **For example**, copper can form ions with a 1+ or 2+ charge, and iron can form ions with a 2+ or 3+ charge.


Periodic Table of the Elements
Period
Group
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
Li+
Be2+
Na+
Mg2+
K+
Ca2+
Rb+
Sr2+
Cs+
Ba2+
Fr+
Ra2+
Cr3+
Cr6+
Mn2+
Fe2+
Fe3+
Co2+
Ni2+
Cu+
Cu2+
Zn2+
Ag+
Cd2+
Pt2+
Au+
Au3+
Hg22+
Hg2+
Al3+
C4−
N3−
P3−
O2−
S2−
Se2−
Te2−
F−
Cl−
Br−
I−
At−
He
Ne
Ar
Kr
Xe
Rn
*
**

- The ions that we have discussed so far are called **monatomic ions**, that is, they are ions formed from only one atom.
- **Polyatomic ions**: are electrically charged molecules (a group of bonded atoms with an overall charge).

  Example: Oxyanions are polyatomic ions that contain one or more oxygen atoms.

Table 2.4: Common Polyatomic Ions

Name	Formula	Related Acid	Formula
Ammonium	NH_4^+		
Hydronium	H_3O^+		
Peroxide	O_2^{2-}		
Hydroxide	OH^-		
Acetate	CH_3COO^-	acetic acid	CH_3COOH
Cyanide	CN^-	hydrocyanic acid	HCN
Azide	N_3^-	hydrazoic acid	HN_3
Carbonate	CO_3^{2-}	carbonic acid	H_2CO_3
Bicarbonate	HCO_3^-		
Nitrate	NO_3^-	nitric acid	HNO_3
Nitrite	NO_2^-	nitrous acid	HNO_2
Sulfate	SO_4^{2-}	sulfuric acid	H_2SO_4

- Note that there is a system for naming some polyatomic ions; **-ate** and **-ite** are **suffixes** designating polyatomic ions **containing more or fewer oxygen atoms**.

- **Per-** (short for "hyper") and **hypo-** (meaning "under") are **prefixes** meaning **more oxygen atoms than -ate** and **fewer oxygen atoms than -ite**, respectively. **For example**, perchlorate is ClO_4^-, chlorate is ClO_3^-, chlorite is ClO_2^- and hypochlorite is ClO^-.

- The number of oxygen atoms corresponding to a given suffix or prefix **is not consistent**; for example**, nitrate is NO_3^-** while **sulfate is SO_4^{2-}**.

2.4.1. Formation of Ionic Compounds

- An ionic compound is formed by the **complete transfer of electrons** from a metal to a nonmetal and the resulting ions have achieved an octet.
- Electrons are actually transferred from one atom to another **to form rare gas electron structures for each ion**. The atom which forms a positive ion loses electrons to the atom which gains electrons to form a negative ion.
- The compound formed by this transfer is stabilized by the electrostatic attractions (ionic bonds) between **the ions of opposite charge** present in the compound.

For Example: Each calcium atom (group 2) can give up two electrons and transfer one to each of two chlorine atoms to form $CaCl_2$, which is composed of Ca^{2+} and Cl^- ions in the ratio of one Ca^{2+} ion to two Cl^- ions.

- **Ionic compounds are solids** that typically **melt at high temperatures** and **boil at even higher temperatures**. For example, sodium chloride melts at 801°C and boils at 1413°C.
- **In solid form**, an ionic compound is not electrically conductive because its ions are unable to flow. **When molten**, however, it can conduct electricity because its ions are able to move freely through the liquid.
- In every ionic compound, the total number of positive charges of the cations equals the total number of negative charges of the anions.

For example, the formula for calcium phosphate is $Ca_3(PO_4)_2$

2.4.2. Formation of Molecular Compounds

- Molecular compounds (covalent compounds) result when atoms share electrons.

- Under normal conditions, molecular compounds often exist as gases, low-boiling liquids, and low-melting solids, although many important exceptions exist.

2.5. Chemical Nomenclature

2.5.1. Ionic compounds

2.5.1.1. Compounds Containing Only Monatomic Ions

- The name of **a binary compound** containing monatomic ions consists of the name of the cation (the name of the metal) followed by the name of the anion (the name of the nonmetallic element with its ending replaced by the suffix -ide).

NaCl, sodium chloride	Na_2O, sodium oxide
KBr, potassium bromide	CdS, cadmium sulfide
CaI_2, calcium iodide	Mg_3N_2, magnesium nitride
CsF, cesium fluoride	Ca_3P_2, calcium phosphide
LiCl, lithium chloride	Al_4C_3, aluminum carbide

2.5.1.2. Compounds Containing Polyatomic Ions

- Compounds containing polyatomic ions are named similarly to those containing only monatomic ions, i.e. by naming first the cation and then the anion

$KC_2H_3O_2$, potassium acetate	NH_4Cl, ammonium chloride
$NaHCO_3$, sodium bicarbonate	$CaSO_4$, calcium sulfate
$Al_2(CO_3)_3$, aluminum carbonate	$Mg_3(PO_4)_2$, magnesium phosphate

2.5.1.3. Compounds Containing a Metal Ion with a Variable Charge

- Most of the transition metals and some main group metals can **form two or more cations with different charges**.

- Compounds of these metals with nonmetals are named by specifying the charge of the metal ion in a **Roman numeral in parentheses** after the name of the metal.

For example, ionic compounds of **iron and chlorine**, $FeCl_2$ and $FeCl_3$. These two compounds are named iron(II) chloride and iron(III) chloride, respectively. Other examples

Compound	Name
$FeCl_2$	iron(II) chloride
$FeCl_3$	iron(III) chloride
Hg_2O	mercury(I) oxide
HgO	mercury(II) oxide
SnF_2	tin(II) fluoride
SnF_4	tin(IV) fluoride

2.5.1.4. Ionic Hydrates

- **Hydrates** are ionic compounds that contain water molecules as integral components of their crystals.

- Ionic compounds with no water are called anhydrous (meaning “not hydrated”)

For example: the anhydrous $CuSO_4$ (copper(II) sulfate) and $CuSO_4 \cdot 5H_2O$ (copper(II) sulfate pentahydrate)
Sodium carbonate decahydrate: $Na_2CO_3 \cdot 10H_2O$

2.5.2. Molecular Compounds

2.5.2.1. Compounds Composed of Two Elements

- When two nonmetallic elements form a molecular compound, several combination ratios are often possible.

For example, carbon and oxygen can form the compounds CO and CO_2. We add prefixes (*mono* and d*i*) to specify the numbers of atoms of each element.

- The name of the more metallic element (the one farther to the left and/or bottom of the periodic table) is first, followed by the name of the more nonmetallic element (the one farther to the right and/or top) with its ending changed to the suffix –ide.

Compound	Name		Compound	Name
SO_2	sulfur dioxide		BCl_3	boron trichloride
SO_3	sulfur trioxide		SF_6	sulfur hexafluoride
NO_2	nitrogen dioxide		PF_5	phosphorus pentafluoride
N_2O_4	dinitrogen tetroxide		P_4O_{10}	tetraphosphorus decaoxide
N_2O_5	dinitrogen pentoxide		IF_7	iodine heptafluoride

2.5.2.2. Binary Acids

- Some compounds containing hydrogen are members of an important class of substances known as acids.
- They **release hydrogen ions**, H^+, when dissolved in water.
- If the compound is a binary acid (comprised of hydrogen and one other nonmetallic element):
  1. The word "hydrogen" is changed to the prefix hydro-
  2. The other nonmetallic element name is modified by adding the suffix -ic
  3. The word "acid" is added as a second word

For example, when the gas HCl (hydrogen chloride) is dissolved in water, the solution is called hydrochloric acid.

Name of Gas	Name of Acid
HF(*g*), hydrogen fluoride	HF(*aq*), hydrofluoric acid
HCl(*g*), hydrogen chloride	HCl(*aq*), hydrochloric acid
HBr(*g*), hydrogen bromide	HBr(*aq*), hydrobromic acid
HI(*g*), hydrogen iodide	HI(*aq*), hydroiodic acid
H_2S(*g*), hydrogen sulfide	H_2S(*aq*), hydrosulfuric acid

2.5.2.3. Oxyacids

- Oxyacids, compounds that contain hydrogen, oxygen, and at least one other element, and are bonded in such a way as to impart acidic properties to the compound.
- Typical oxyacids consist of hydrogen combined with a polyatomic, oxygen-containing ion. To name oxyacids:
  1. Omit “hydrogen”
  2. Start with the root name of the anion
  3. Replace –ate with –ic, or –ite with –ous
  4. Add “acid”

Formula	Anion Name	Acid Name
$HC_2H_3O_2$	Acetate	acetic acid
HNO_3	Nitrate	nitric acid
HNO_2	Nitrite	nitrous acid
$HClO_4$	Perchlorate	perchloric acid
H_2CO_3	Carbonate	carbonic acid
H_2SO_4	Sulfate	sulfuric acid
H_2SO_3	Sulfite	sulfurous acid
H_3PO_4	Phosphate	phosphoric acid